

Freight Rate Prediction

End-to-End Machine Learning Engineering Report

A reproducible temporal tabular regression solution using leakage-safe preprocessing, engineered route features, expanding-window validation, and CatBoost.

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REPORT DATE	29 August 2026

\$100.99 Mean forward-validation MAE Tuned model • 3 future folds	38 Final engineered features Coordinates + load + calendar	PASSED Provided scorer validation 12,000 + 31 rows validated
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Submission status

The model pipeline, final validation predictions, fixed December predictions, saved model artifacts, and scorer-generated December chart are complete. Hidden final validation metrics are calculated by Spotter after submission.

Executive Summary

This assessment develops a production-oriented freight-rate regression pipeline from the provided January–October 2025 labeled loads and applies it to 12,000 unlabeled November–December loads. The solution is deliberately organized around future-period generalization: every modeling decision is tested with expanding-window chronological validation rather than a random split.

48,000 Labeled development rows Jan–Oct 2025	12,000 Final validation rows Nov–Dec 2025	3 Chronological holdout folds Aug, Sep, Oct
1,557 Final boosting rounds Derived from fold optima	0.8277 Mean internal R² Selected tuned model	\$858.78 December mean prediction Fixed 360-mile route

Key conclusions

- Distance is the dominant raw predictor (Pearson $r \approx 0.909$ with `posted_rate`), while raw weight and the supplied market signals have weak marginal relationships with the target.
- Negative weights behave like sign-entry errors: their absolute magnitudes match the valid positive-weight distribution. They are corrected with absolute value rather than discarded.
- Eight city names appear in final validation but not in labeled development data, affecting 1,447 validation rows (12.06%). Coordinate-derived route features therefore provide safer geographic generalization than relying on raw city labels.
- Feature Set B — coordinates and engineered route/calendar/load features, without `market_index`, `quote_signal`, `pickup`, or `delivery` — produced the best initial forward-validation MAE (\$103.29).
- Controlled tuning selected `T4_depth8_regularized` and improved mean forward-validation MAE to \$100.99, with worst-fold MAE \$107.46.
- Keeping all target observations outperformed removing strong residual anomalies and narrowly outperformed target capping, so no target labels are deleted or altered.
- The final model is `CatBoostRegressor` trained on all 48,000 labeled rows with 38 features and 1,557 boosting rounds. Both required prediction files pass the supplied `score.py` validator.

Most important methodological choice

The assessment is treated as temporal tabular regression, not classical time-series forecasting. Individual load characteristics drive each prediction, while chronology is used to construct realistic future holdouts and to prevent leakage.

Report Map

The report is arranged around the evidence → decision → validation chain so that each modeling choice can be traced to a measured data property or controlled experiment.

Section	What it answers
1. Assessment Objective & Success Criteria	What must be predicted, delivered, and validated?
2. Data Understanding & EDA	What does the data contain, and what problems or patterns matter?
3. Preprocessing & Feature Engineering	How are data issues corrected without leakage, and why are features constructed this way?
4. Validation Design	How is future generalization measured realistically?
5. Model Development & Selection	Why CatBoost, which feature set, which hyperparameters, and how are anomalies handled?
6. Final Training & Predictions	How is the final model trained, saved, and used for the two required prediction outputs?
7. Error Analysis & Interpretation	What do internal errors say about model strengths and remaining risk?
8. Reproducibility & Submission Readiness	Can the result be rerun and does it satisfy the supplied scorer?
9. Limitations & Future Improvements	What is not proven by the current evidence?
Appendices	Exact final features, commands, and artifact inventory.

End-to-end assessment workflow



All fitted transformations are learned from training data only; final validation labels remain unseen.

Figure 1. End-to-end assessment workflow.

1. Assessment Objective and Success Criteria

Objective: Predict posted freight rates for every row in validation.csv and produce the fixed 31-day December prediction file required by the supplied scorer. The final solution must be reproducible, preserve the provided source files, and generate positive finite predictions in the exact required formats.

Deliverable	Requirement	Completed result
Final predictions	validation Exactly 12,000 rows; columns load_id,predicted_rate; IDs TE-000001...TE-012000; positive finite predictions	validation_predictions.csv created and scorer-validated
Fixed predictions	December 31 daily rows for 2025-12-01...2025-12-31; Lexington → Fort Wayne; 360 miles; Dry Van; 32,000 lb	data/december_chart_inputs.csv created and scorer-validated
December chart	Generated by the supplied score.py	scorer_results/candidate_december.png created
Model artifact	Reusable trained estimator plus transformation metadata	CatBoost .cbm model + preprocessing, feature, and configuration JSON files saved
Performance reporting	Internal validation before hidden scoring	3 expanding-window folds; final hidden metrics left to Spotter

1.1 Problem framing

Each row is an independent freight load with route, equipment, weight, date, and contextual variables. The target is a continuous dollar rate. This is therefore supervised tabular regression with a temporal deployment boundary. A classical time-series model would underuse the load-level route and equipment information; a random split would overstate generalization by allowing later-period observations into training.

Decision

Use tabular regression for the estimator and chronology for validation. This matches the actual submission condition: train on historical labeled loads, then predict later unseen loads.

2. Data Understanding and Exploratory Analysis

2.1 Dataset inventory

Dataset	Rows	Columns	Date coverage	Role
train-test.csv	48,000	14	2025-01-01 → 2025-10-31	Labeled development data
validation.csv	12,000	13	2025-11-01 → 2025-12-31	Unlabeled final predictions

Dataset	Rows	Columns	Date coverage	Role
validation-predictions-template.csv	12,000	2	n/a	Required output order/schema
december-chart-inputs.csv	31	7	2025-12-01 → 2025-12-31	Fixed scenario for scorer chart

Structural audit: No duplicate rows or duplicate load IDs were found in the labeled or final-validation data. Dates parse cleanly, distances are positive, coordinates lie in valid ranges, and equipment categories are consistent between development and validation.

2.2 Target behavior

Posted_rate is right-skewed: the median is \$2,030.76, the mean is \$2,373.98, the 99th percentile is \$5,972.83, and the maximum is \$25,533. The long upper tail is material for model design because squared-error objectives would give these rare observations disproportionate influence.

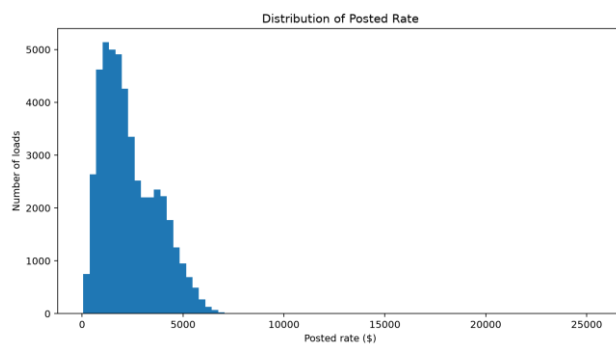


Figure 2a. Posted-rate distribution.

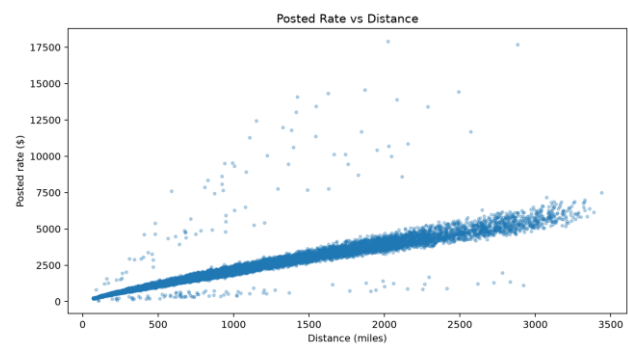


Figure 2b. Posted rate versus distance.

Distance is the strongest raw numerical relationship with the target (Pearson $r \approx 0.909$). The scatter plot shows a dense, nearly linear dominant band with a much smaller set of unusually high or low rates. This motivates using distance directly while also giving the model nonlinear route features and a robust MAE objective.

Decision logic

Keep the original dollar target rather than log-transforming it for the primary model, and use MAE as CatBoost's loss/evaluation metric. MAE preserves direct business interpretation in dollars and is less dominated by the rare extreme rates than squared-error loss.

2.3 Data-quality anomalies

Issue	Train	Validation	Interpretation
Missing weight	300 (0.62%)	165 (1.38%)	Sparse; impute after fitting statistics on training only

Issue	Train	Validation	Interpretation
Negative weight	292 (0.61%)	145 (1.21%)	Physically invalid sign, but magnitude distribution is plausible
Missing market_index	374 (0.78%)	249 (2.08%)	Sparse; initially impute for feature-set experiments
Invalid/non-positive distance	0	0	No correction required
Invalid date	0	0	No correction required

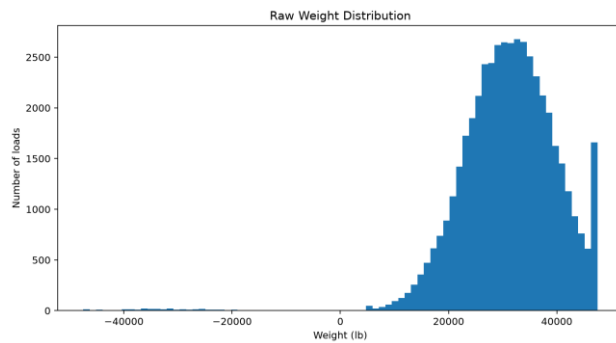


Figure 3a. Raw weight exposes negative-value sign errors.

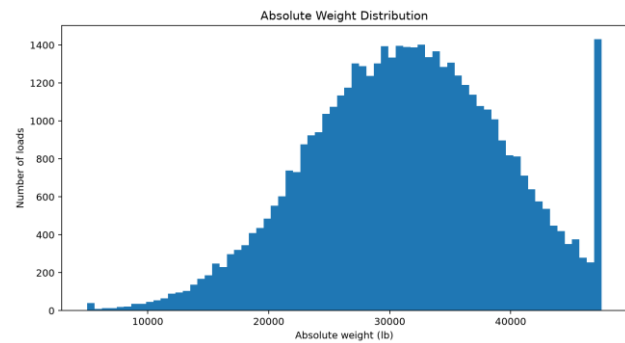


Figure 3b. Absolute weight restores the plausible physical distribution.

The negative-weight records are not random gross outliers: after taking absolute values, their magnitude distribution aligns closely with valid positive loads. Negative rows have median $|\text{weight}| \approx 31,822$ lb versus an overall positive median around 31,494 lb. Deleting them would discard otherwise useful route and target information.

Decision

Convert negative weight to $\text{abs}(\text{weight})$, retain a `weight_negative_flag`, and add a separate `weight_missing_flag`. Missing weights are filled with equipment-specific medians learned from the current training fold, with the global training median as fallback.

2.4 Equipment, geography, and temporal behavior

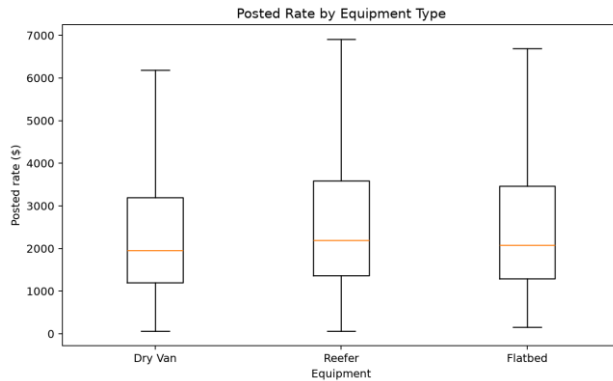


Figure 4a. Posted-rate distribution by equipment type.

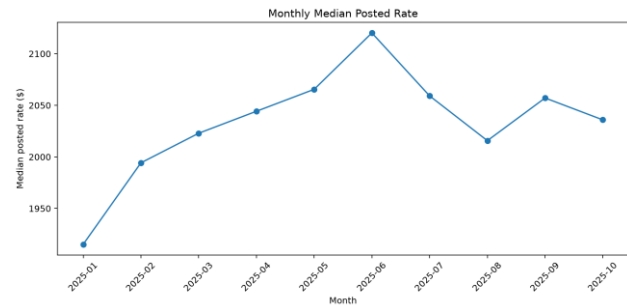


Figure 4b. Monthly median posted rate.

All three equipment classes — Dry Van, Reefer, and Flatbed — are represented in both development and final validation. Median rates differ by equipment (approximately \$1,953 Dry Van, \$2,197 Reefer, and \$2,077 Flatbed), so equipment is retained as CatBoost's categorical feature.

Monthly median rates move from \$1,915 in January to a local peak of \$2,120 in June, then end October near \$2,036. Mean distance remains comparatively stable around 1,130–1,145 miles across months, indicating that temporal rate movement is not explained solely by distance mix.

The geographic audit identified eight validation cities absent from the corresponding labeled city categories: Allentown, Charlotte, Chicago, Jackson, Knoxville, Laredo, Norfolk, and San Diego. At least one unseen pickup or delivery occurs in 1,447 validation rows (12.06%).

Decision

Do not make raw pickup/delivery city labels part of the final feature set unless forward validation proves they help. Instead, emphasize latitude/longitude and route geometry so the model can generalize to unseen city names.

2.5 Development-to-validation distribution check

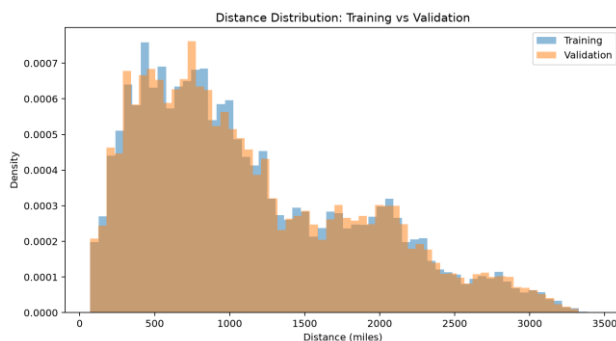


Figure 5a. Distance distribution: development versus final validation.

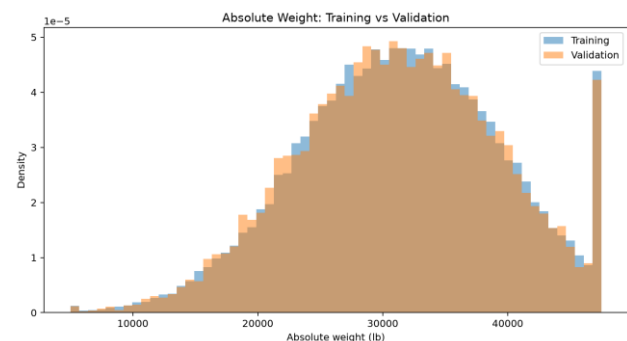


Figure 5b. Absolute-weight distribution: development versus final validation.

Distance and absolute-weight distributions overlap closely between development and final validation, which reduces concern about broad covariate shift in the two strongest physical dimensions. The city-category coverage difference remains the more important deployment risk.

3. Preprocessing and Feature Engineering

3.1 Leakage-safe preprocessing architecture

Preprocessing is implemented as two operations: `fit_preprocessing()` learns statistics and `apply_preprocessing()` applies them. During every chronological fold, fitting occurs on the fold's training portion only. No medians, thresholds, or other learned transformation parameters are estimated from the future holdout or from final validation.

Transformation	Implementation	Why this choice
Negative weight	<code>weight_clean = abs(weight); add weight_negative_flag</code>	Corrects an impossible sign without discarding an otherwise plausible load
Missing weight	Equipment median → global median fallback; add <code>weight_missing_flag</code>	Uses local load context while preserving a missingness signal
Missing market_index	Training median + <code>market_index_missing_flag</code>	Needed for fair initial feature-set experiments; later excluded from final feature set
Date	Strict datetime parsing; reject invalid values	Enables chronological splits and calendar features
Target anomalies	No deletion/capping during preprocessing	Whether to alter labels is treated as an empirical model-selection question, not a cleaning assumption
Raw files	Never overwritten	Preserves provenance and makes the pipeline repeatable

3.2 Preprocessing verification

A dedicated verification script confirmed that the transformation removes all missing/negative values that the model depends on: training missing weight 300 → 0, negative weight 292 → 0, missing market_index 374 → 0; validation missing weight 165 → 0, negative weight 145 → 0, and missing market_index 249 → 0. Dates remain fully valid.

Why no permanent cleaned training CSV?

A single pre-cleaned file would encourage fitting imputation values on all labeled data before internal validation. Transforming in memory inside each fold guarantees that every holdout remains genuinely future/unseen.

3.3 Feature engineering

The engineered feature space combines raw physical variables with route geometry, calendar position, and interaction terms. The final selected configuration contains 38 features and only one categorical feature (equipment).

Feature family	Examples	Purpose
Physical load	distance, log_distance, weight_clean, weight flags	Represent shipment scale and known data-quality state
Route coordinates	pickup/delivery latitude & longitude	Preserve geography without depending on city-name memorization
Route geometry	lat/lon change, midpoint, haversine distance, detour ratio, bearing sin/cos	Capture direction, straight-line separation, and route efficiency
Calendar	month, day of week/year, week, quarter, weekend/month-boundary flags	Represent temporal market patterns
Cyclical time	sin/cos for day-of-week, month, day-of-year	Respect periodic proximity (e.g., Sunday – Monday; Dec – Jan)
Interactions	distance × weight, weight-per-distance proxy	Let load size modify distance-related rate behavior
Categorical	equipment	Capture equipment-specific rate structure

3.4 Controlled feature configurations

Rather than assuming that every available field helps, three feature configurations were built and compared under identical CatBoost settings. This converts feature inclusion into a measured decision.

Set	Features	Purpose of experiment
A	41 features: coordinates + market_index + quote_signal	Test whether supplied market signals improve future generalization
B	38 features: coordinates, no market signals, no raw city labels	Coordinate-led generalization with fewer potentially unstable signals
C	43 features: A + pickup + delivery categories	Test whether CatBoost city categories add value despite unseen final-validation cities

4. Validation Design

4.1 Why random splitting was rejected

The labeled data end in October while the final validation period begins in November. A random split would allow later-season observations into training while evaluating earlier dates, creating a validation setting that is easier than deployment. The solution therefore uses expanding-window future holdouts.

Fold	Training window	Train rows	Holdout	Holdout rows
1	Jan 1 – Jul 31	33,718	August	4,759
2	Jan 1 – Aug 31	38,477	September	4,670
3	Jan 1 – Sep 30	43,147	October	4,853

Each fold passed two explicit integrity checks: $\max(\text{training date}) < \min(\text{holdout date})$, and load IDs are disjoint. Preprocessing statistics are refitted independently within each fold.

Selection metric

Mean MAE across the three future holdouts is the primary model-selection criterion. Worst-fold MAE is tracked as a stability measure, while RMSE and R^2 provide complementary views of tail error and explained variance.

5. Model Development and Selection

5.1 Why CatBoost

CatBoost was selected as the primary model family because the problem is medium-sized tabular regression with nonlinear numeric relationships, one important categorical variable, interaction effects, and rare target extremes. It handles categorical features natively, does not require feature scaling, models nonlinear interactions efficiently, and supports MAE loss plus early stopping.

The assessment deliberately focuses experimentation on feature representation, temporal validation, regularization, and robustness rather than building a broad model zoo. Consequently, the report claims CatBoost is the best tested configuration here — not that it is universally superior to every possible algorithm.

5.2 Feature-set selection

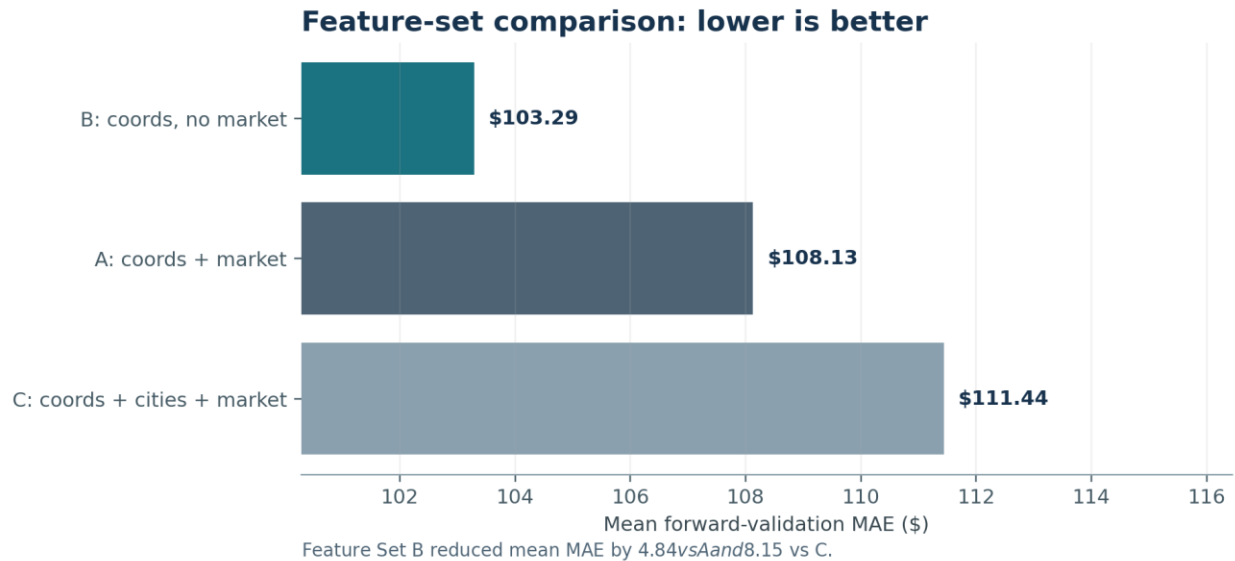


Figure 6. Initial feature-set comparison under identical CatBoost settings.

Feature set	Mean MAE	Mean RMSE	Mean R ²	Decision
B – coordinates, no market	\$103.29	\$626.47	0.8276	Selected
A – coordinates + market	\$108.13	\$627.64	0.8269	Rejected
C – coordinates + cities + market	\$111.44	\$629.54	0.8259	Rejected

Feature Set B improves mean MAE by \$4.84 versus A and \$8.15 versus C. The result also aligns with the EDA: raw market signals show weak marginal target relationships, while city names face unseen-category risk in final validation.

Decision

Lock Feature Set B for all later tuning. This keeps the feature representation fixed so hyperparameter experiments answer one question at a time.

5.3 Controlled CatBoost tuning

Six deliberately different parameter sets tested tree depth, learning rate, L2 regularization, and random strength. Each configuration used up to 2,500 boosting rounds with MAE evaluation and 150-round early stopping. `use_best_model=True` retained the best iteration within each fold.

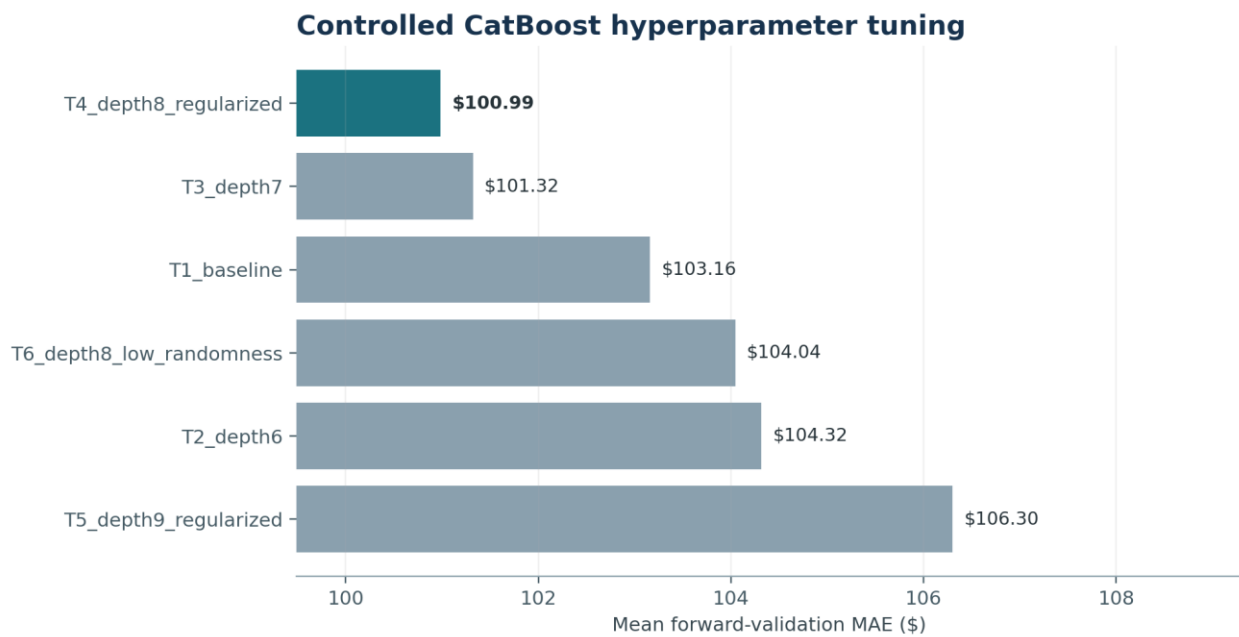


Figure 7. Mean forward-validation MAE across six CatBoost configurations.

Configuration	Depth	LR	L2	Mean MAE	Worst-fold MAE	Mean best iteration
T4 depth8 regularized	8	0.025	10	\$100.99	\$107.46	1,556.3
T3 depth7	7	0.035	5	\$101.32	\$111.45	1,746.0
T1 baseline	8	0.030	5	\$103.16	\$114.24	1,132.7
T6 depth8 low randomness	8	0.030	7	\$104.04	\$112.85	1,273.0
T2 depth6	6	0.040	5	\$104.32	\$118.27	1,270.0
T5 depth9 regularized	9	0.025	10	\$106.30	\$114.02	1,002.7

T4_depth8_regularized provides the lowest mean MAE and a strong worst-fold result. The smaller learning rate and stronger L2 penalty improve average generalization without materially changing RMSE, indicating that tuning mainly reduces the typical error rather than eliminating rare large misses.

Decision

Select depth 8, learning rate 0.025, L2 leaf regularization 10, random strength 1.0, and MAE loss. The final iteration count is set to 1,557, obtained by rounding the mean fold best-iteration index and converting the zero-based index into a tree count.

5.4 Target-anomaly robustness test

Target outliers were not automatically treated as bad data. After tuning, three strategies were tested with the same feature set and CatBoost parameters: keep all targets unchanged; remove strong residual anomalies identified only within the training fold; or cap the lower/upper 0.5% target tails within the training fold.

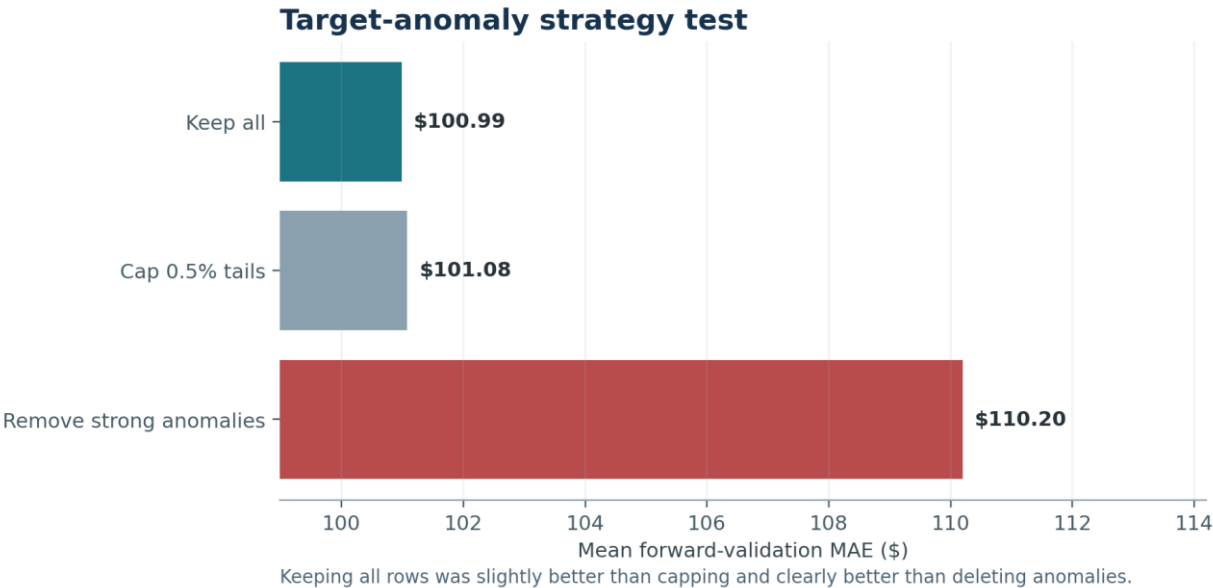


Figure 8. Target-handling strategy comparison.

Strategy	Mean MAE	Worst-fold MAE	Mean RMSE	Outcome
Keep all	\$100.99	\$107.46	\$626.25	Selected
Cap 0.5% tails	\$101.08	\$107.35	\$626.26	No improvement
Remove strong anomalies	\$110.20	\$126.16	\$628.95	Materially worse

Deleting strong anomalies sharply worsens October MAE, which suggests that at least some extreme labels represent legitimate freight-rate behavior rather than simple corruption. Capping is nearly neutral but still slightly worse on average.

Decision

Keep every labeled row and preserve original target values. Robustness is provided by MAE training rather than destructive label cleaning.

5.5 Final selected model

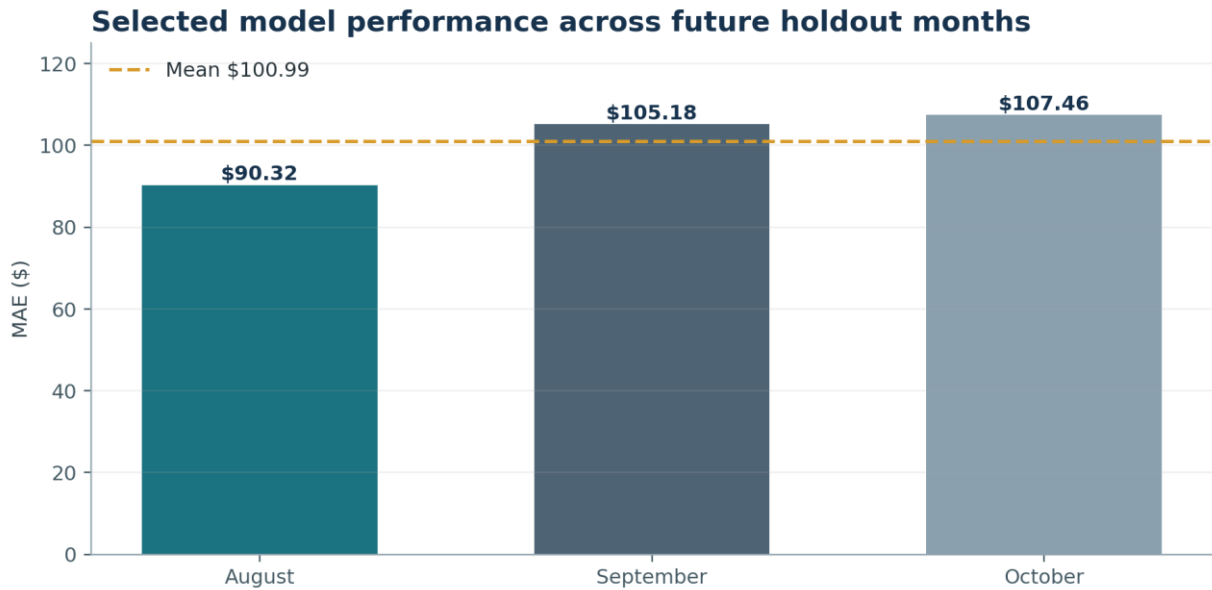


Figure 9. Selected T4 model MAE across the three chronological holdout months.

Component	Final choice
Estimator	CatBoostRegressor
Feature set	B_coordinates_no_market — 38 features; equipment categorical
Target handling	Keep all 48,000 labels unchanged
Loss / selection metric	MAE
Depth / learning rate	8 / 0.025
L2 / random strength	10 / 1.0
Final boosting rounds	1,557
Random seed	42
Internal mean metrics	MAE \$100.99 • RMSE \$626.25 • R² 0.8277
Internal worst-fold MAE	\$107.46

6. Final Training and Prediction Outputs

6.1 Full-data final training

After all choices were locked, preprocessing statistics were fitted once on the complete 48,000-row labeled dataset, Feature Set B was generated, and the selected CatBoost configuration was trained for 1,557 rounds without a validation set. This is appropriate because all labeled data can now contribute to the deployable model; the iteration count was already chosen from internal future folds.

Saved artifact	Purpose
models/final_catboost_model.cbm	Serialized final CatBoost estimator
models/preprocessing_stats.json	Equipment-specific weight medians, global weight median, and compatibility market median
models/feature_metadata.json	Exact 38 feature columns, order, options, and categorical definition
models/final_model_config.json	Selected hyperparameters, target-handling rule, seed, and feature-set ID
reports/tables/final_model_training_summary.csv	Compact machine-readable training record

Why save metadata as well as model weights?

The .cbm file alone cannot reproduce the exact input matrix. Prediction scripts therefore load the same preprocessing statistics, feature options, column order, and categorical definition that were used for final training.

6.2 Final validation predictions

The final model produced 12,000 positive finite rates and merged them into the supplied prediction template by load_id rather than relying on row order. The output contains exactly the two required columns and IDs TE-000001 through TE-012000.

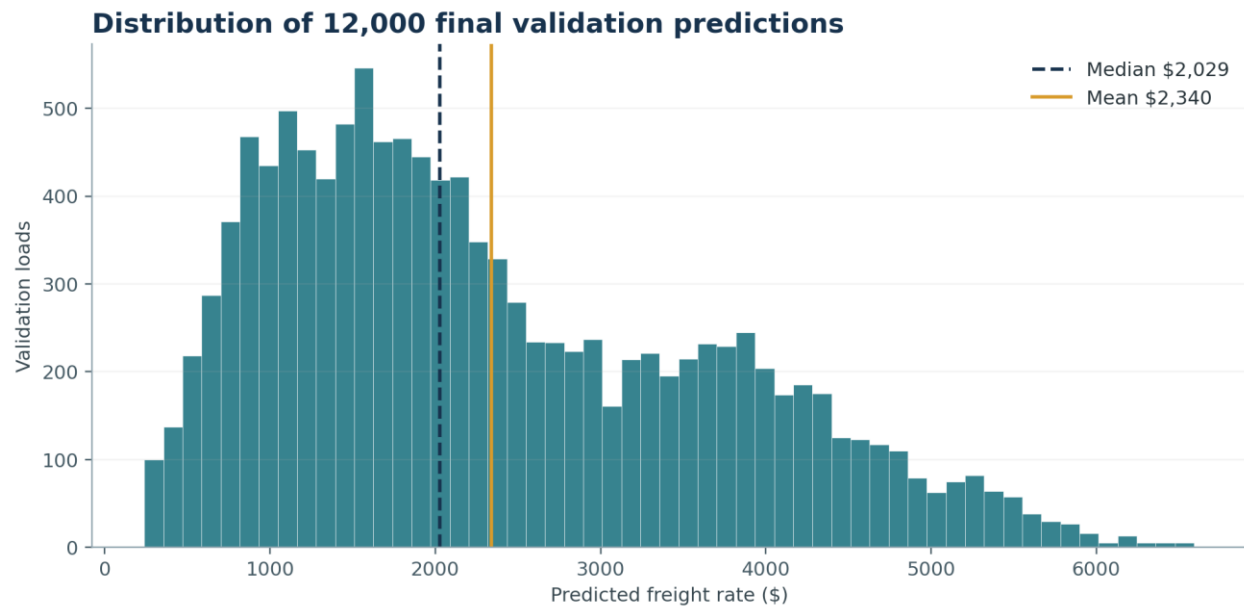


Figure 10. Distribution of final validation predictions.

Statistic	Predicted rate
Count	12,000
Minimum	\$239.94
25th percentile	\$1,289.76
Median	\$2,029.29
Mean	\$2,339.60
75th percentile	\$3,295.51
Maximum	\$6,590.17

The median final prediction (\$2,029.29) is very close to the labeled target median (\$2,030.76), while the predicted maximum is far below the rare training maximum of \$25,533. This is consistent with an MAE-trained ensemble that favors stable central predictions and does not chase a small number of extreme labels.

6.3 Fixed December prediction workflow

The supplied December file contains pickup, delivery, distance, equipment, weight, date, and an empty predicted_rate column, but not the coordinate fields expected by Feature Set B. To keep the source file unchanged and maintain model consistency, the prediction script creates an in-memory copy, derives Lexington and Fort Wayne coordinates from the labeled training city-coordinate mapping, and then builds the same 38 features used in final training.

market_index is absent from the December input. Because the final Feature Set B excludes market_index and quote_signal, market information has no influence on December predictions. A temporary missing market_index field is created only to satisfy the shared preprocessing function; it is not part of the final model matrix or output.

Fixed input	Value
Pickup	Lexington
Delivery	Fort Wayne
Distance	360 miles
Equipment	Dry Van
Weight	32,000 lb
Dates	December 1–31, 2025
Only changing model input	Date-derived features

Candidate: December 2025 Predicted Load Rate

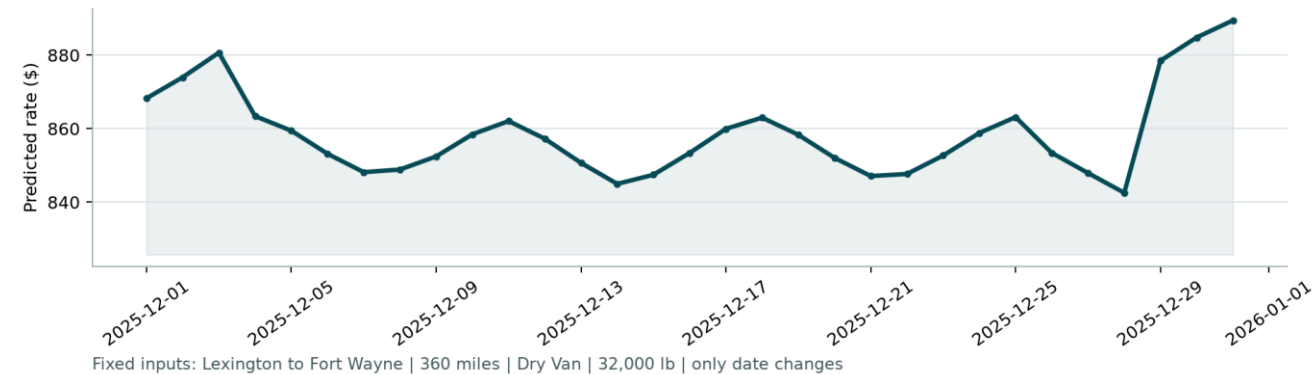


Figure 11. Scorer-generated December 2025 predicted load-rate chart.

December statistic	Predicted rate
Minimum	\$842.56
Mean	\$858.78
Median	\$857.27
Maximum	\$889.45
Standard deviation	\$11.82

The narrow range is expected: route, distance, equipment, and weight are fixed across all 31 rows, so only date-derived features vary. The pattern should therefore be interpreted as the model's learned calendar adjustment for an otherwise identical load, not as a full market forecast.

6.4 Supplied scorer result

SCORER VALIDATION PASSED

score.py validated all 12,000 final predictions, validated all 31 fixed December predictions, and created scorer_results/candidate_december.png. The scorer explicitly states that final validation metrics are calculated by Spotter after submission.

This confirms schema, row count, ID coverage, positive numeric predictions, exact December fixed inputs, complete December dates, and successful chart generation. It does not reveal hidden validation accuracy, so internal expanding-window metrics remain the only available pre-submission estimate of predictive performance.

7. Error Analysis and Interpretation

7.1 Fold-level behavior

Holdout month	MAE	RMSE	R ²	Interpretation
August	\$90.32	\$613.78	0.8268	Best typical error; still exposed to rare large misses
September	\$105.18	\$616.53	0.8362	Highest explained variance; moderate MAE
October	\$107.46	\$648.45	0.8200	Hardest fold; selected as worst-fold reference

The mean MAE of \$100.99 is much smaller than mean RMSE of \$626.25. This gap is not contradictory: MAE describes the typical dollar miss, whereas RMSE disproportionately reflects the small number of extreme rate errors. The earlier residual audit found 664 strong target-anomaly candidates and examples with residual magnitudes above \$20,000 under a simple diagnostic baseline.

Because removing these rows worsened chronological validation, the final interpretation is that tail behavior is partly real and difficult rather than simply dirty. The selected model prioritizes stable overall accuracy while accepting that extreme freight rates remain the main source of squared-error risk.

7.2 What the internal metrics do and do not prove

- They estimate performance on future months drawn from the same 2025 development process.
- They show that the chosen feature representation and tuning decisions improve average future-period MAE under controlled comparisons.
- They do not provide the true error on the 12,000 final validation rows because their labels are withheld.
- They do not prove CatBoost is globally optimal versus all other model families; alternative ensembles remain a potential extension.

Practical expectation

The final submission should be judged primarily against Spotter's hidden labels. The internal MAE of approximately \$101 is a disciplined validation estimate, not a claimed hidden-test score.

8. Reproducibility and Submission Readiness

8.1 Environment and dependency control

Development was performed in Visual Studio Code with a dedicated Python virtual environment. The project requirements constrain matplotlib $\geq 3.8, < 4$; numpy $\geq 1.26, < 3$; pandas $\geq 2.0, < 3$; scikit-learn $\geq 1.4, < 2$; and catboost $\geq 1.2, < 2$. The recorded environment used Python 3.14.3.

8.2 Reproducible pipeline

Stage	Script	Primary artifact
Data audit	src/data_audit.py	reports/tables/* audit CSVs
EDA	src/eda.py	23 figures + EDA summary tables
Anomaly investigation	src/anomaly_analysis.py; src/residual_analysis.py	anomaly and residual diagnostic tables
Preprocessing verification	src/preprocessing_check.py	preprocessing_check_summary.csv
Chronological verification	split src/split_check.py	chronological_split_summary.csv
Feature verification	src/features_check.py	feature_inventory.csv
Feature-set comparison	src/train_catboost.py	catboost_feature_set_comparison.csv
Hyperparameter tuning	src/catboost_tuning.py	catboost_tuning_summary.csv
Anomaly strategy test	src/catboost_anomaly_test.py	catboost_anomaly_strategy_comparison.csv
Final training	src/train_final_model.py	model .cbm + JSON metadata
Final validation prediction	src/predict_validation.py	validation_predictions.csv
December prediction	src/predict_december.py	data/december_chart_inputs.csv
Scorer	score.py	scorer_results/candidate_december.png

8.3 Submission readiness checklist

Check	Status
Original assessment files preserved under data/raw	PASS

Check	Status
48,000-row final model trained and serialized	PASS
12,000 validation predictions generated	PASS
validation_predictions.csv exact schema and IDs	PASS
31 fixed December predictions generated	PASS
December original seven-column schema preserved	PASS
Provided score.py executed successfully	PASS
candidate_december.png generated	PASS
Final hidden metrics available locally	NO — calculated by Spotter after submission

9. Limitations and Potential Improvements

- Hidden validation labels are unavailable; the report cannot state final Spotter MAE/RMSE/R² before submission.
- The model family search is intentionally focused on CatBoost. A future benchmark could compare LightGBM/XGBoost, Extra Trees, or calibrated ensembles under the same chronological folds.
- Rare rate extremes dominate RMSE and remain difficult. Future work could explore specialized tail models, mixture-of-experts approaches, quantile objectives, or separate anomaly-regime detection — but only if validated chronologically.
- The final feature set excludes market_index and quote_signal because they worsened the tested forward-validation configuration. This is an empirical result for this dataset, not a claim that market signals are universally unhelpful.
- City categories were excluded because they worsened tested performance and final validation contains unseen cities. More sophisticated geographic encodings (regions, lane clusters, learned embeddings) could be evaluated.
- December predictions depend on date-derived variation for a fixed lane. Because the assessment file omits coordinates and market fields, coordinates are reconstructed from the training city mapping; this should be documented in any production deployment.

9.1 Recommended next experiments

Priority	Experiment	Success criterion
1	Gradient-boosting ensemble under identical time folds	Lower mean MAE without worse worst-fold MAE
2	Route-cluster / geospatial encoding	Improved unseen-city generalization

Priority	Experiment	Success criterion
3	Tail-aware objective or two-stage model	Lower RMSE while retaining MAE $\approx$ \$101 or better
4	Feature importance + SHAP audit	Confirm model reliance aligns with business logic and detect unstable shortcuts
5	Model monitoring design	Track drift in lane mix, equipment, weight, and predicted-rate distribution after deployment

10. Conclusion

The completed solution is a leakage-aware, reproducible freight-rate regression pipeline designed around the actual future-period deployment condition. EDA established a strong distance-rate relationship, sparse but correctable weight/market missingness, and meaningful geographic generalization risk from unseen city categories. These findings directly shaped preprocessing and feature design.

Controlled forward experiments then selected a 38-feature coordinate-led representation, a regularized depth-8 CatBoost model, and an explicit decision to retain all target observations. The selected model achieves mean internal forward-validation MAE of \$100.99, worst-fold MAE of \$107.46, mean RMSE of \$626.25, and mean R^2 of 0.8277.

The final CatBoost model was trained on all 48,000 labeled loads for 1,557 boosting rounds and saved with the exact preprocessing and feature metadata required for repeatable inference. It generated all 12,000 validation predictions and all 31 fixed December predictions. The supplied scorer accepted both files and generated the required December chart. The assessment is therefore technically complete and ready for submission, with final external validation metrics to be calculated by Spotter.

Final outcome

A complete, auditable machine learning workflow has been delivered: raw-data preservation → evidence-based cleaning → engineered route features → chronological validation → controlled CatBoost selection → robustness testing → saved final model → scorer-validated outputs.

Appendix A. Final 38-Feature Inventory

Feature family	Final features
Physical / load	distance, log_distance, weight_clean, weight_missing_flag, weight_negative_flag, equipment
Coordinates	pickup_lat, pickup_lon, delivery_lat, delivery_lon
Route geometry	latitude_change, longitude_change, absolute_latitude_change, absolute_longitude_change, route_mid_lat, route_mid_lon, haversine_distance, detour_ratio, bearing_sin, bearing_cos
Calendar	year, month, day_of_month, day_of_week, day_of_year, week_of_year, quarter, is_weekend, is_month_start, is_month_end
Cyclical calendar	day_of_week_sin, day_of_week_cos, month_sin, month_cos, day_of_year_sin, day_of_year_cos
Interactions	distance_x_weight, weight_per_1000_miles

Excluded from the final matrix after controlled validation: market_index_clean, market_index_missing_flag, quote_signal, pickup, and delivery. load_id and posted_rate are identifiers/target rather than predictors.

Appendix B. Key Execution Commands

```
python .\src\data_audit.py
python .\src\eda.py
python .\src\anomaly_analysis.py
python .\src\residual_analysis.py
python .\src\preprocessing_check.py
python .\src\split_check.py
python .\src\features_check.py
python .\src\train_catboost.py
python .\src\catboost_tuning.py
python .\src\catboost_anomaly_test.py
python .\src\train_final_model.py
python .\src\predict_validation.py
python .\src\predict_december.py
python .\score.py --predictions .\validation_predictions.csv --december-predictions .\data\december_chart_inputs.csv
```

Appendix C. Principal Generated Artifacts

Artifact	Role
validation_predictions.csv	Final 12,000-row submission predictions
data/december_chart_inputs.csv	Completed fixed December scorer input

Artifact	Role
scorer_results/candidate_december.png	Scorer-generated required chart
models/final_catboost_model.cbm	Final serialized model
models/preprocessing_stats.json	Saved preprocessing parameters
models/feature_metadata.json	Exact feature schema
models/final_model_config.json	Final hyperparameter/configuration record
reports/tables/catboost_tuning_summary.csv	Tuning evidence
reports/tables/catboost_anomaly_strategy_comparison.csv	Target-handling evidence
reports/run_logs/session_01.txt	Execution trace

Appendix D. Additional Exploratory Data Analysis (EDA) Assets

The project also contains 23 EDA figures covering target and rate-per-mile distributions, distance, raw/absolute weight, equipment, monthly/day-of-week behavior, market_index, quote_signal, missingness, and development-versus-validation distributions. Detailed audit, anomaly, residual, preprocessing, split, feature, model, and prediction tables are stored under reports/tables for independent inspection.